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Code | git

```
[*]: utils.get_trained_model()
```

Processing:

Initiated	15:32:02
Status	Preprocessing Data

Following data types have been inferred automatically, if they are correct press enter to continue or type 'quit' otherwise

	Data Type
city_tier	Numeric
first_platform_c	Categorical
first_utm_medium_c	Categorical
first_utm_source_c	Categorical
total_leads_droppped	Numeric
referred_lead	Categorical
app_complete_flag	Label

```
[*]: # setup pycaret
!pip install pycaret
```

4 : Model Experimentation with pycaret

```
[*]: # create a experiaentation with pycaret and exclude ['gbc','knn','qda', 'dummy', 'svm', 'ada']

data_unseen = df.sample(frac=0.1, random_state=42)      # Sample 10% of the data to become the unseen test set
df = df.drop(data_unseen.index)                         # Use the remaining 90% as the training (& validation) data

df.reset_index(drop=True, inplace=True)
data_unseen.reset_index(drop=True, inplace=True)

print('Data for Model Training & Validation: ' + str(df.shape))
print('Unseen Data For Predictions: ' + str(data_unseen.shape))
```

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Name	Last Modified
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logs.log	seconds ago
utils.py	6 hours ago

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Code git

[11]:
```
utils.get_trained_model()
```

Split	Fold	MAE	MSE	RMSE	R2	RMSLE	MAPE
	0	0.3408	0.1700	0.4123	0.3200	0.2892	0.3401
	1	0.3384	0.1680	0.4099	0.3280	0.2874	0.3375
	2	0.3397	0.1690	0.4111	0.3241	0.2882	0.3372
CV-Val	3	0.3382	0.1680	0.4099	0.3279	0.2870	0.3382
	4	0.3388	0.1678	0.4097	0.3287	0.2875	0.3371
	Mean	0.3392	0.1686	0.4106	0.3257	0.2879	0.3380
	Std	0.0009	0.0008	0.0010	0.0033	0.0008	0.0011
Train	nan	0.3384	0.1680	0.4099	0.3280	0.2875	0.3363

```
LGBMRegressor(boosting_type='gbdt', class_weight=None, colsample_bytree=1.0,
              importance_type='split', learning_rate=0.1, max_depth=-1,
              min_child_samples=20, min_child_weight=0.001, min_split_gain=0.0,
              n_estimators=100, n_jobs=-1, num_leaves=31, objective=None,
              random_state=42, reg_alpha=0.0, reg_lambda=0.0, silent='warn',
              subsample=1.0, subsample_for_bin=200000, subsample_freq=0)

LGBMRegressor(bagging_fraction=0.6, bagging_freq=3, boosting_type='gbdt',
              class_weight=None, colsample_bytree=1.0, feature_fraction=0.7,
              importance_type='split', learning_rate=0.2, max_depth=-1,
              min_child_samples=81, min_child_weight=0.001, min_split_gain=0.1,
              n_estimators=90, n_jobs=-1, num_leaves=40, objective=None,
              random_state=42, reg_alpha=0.0005, reg_lambda=0.7, silent='warn',
              subsample=1.0, subsample_for_bin=200000, subsample_freq=0)
```

[12]:
```
# setup pycaret
```

Code

```
[*]: utils.get_trained_model()
```

	Description	Value
0	session_id	42
1	Target	app_complete_flag
2	Original Data	(238964, 7)
3	Missing Values	False
4	Numeric Features	2
5	Categorical Features	4
6	Ordinal Features	False
7	High Cardinality Features	False
8	High Cardinality Method	None
9	Transformed Train Set	(191171, 38)
10	Transformed Test Set	(47793, 38)
11	Shuffle Train-Test	True
12	Stratify Train-Test	False
13	Fold Generator	KFold
14	Fold Number	10
15	CPU Jobs	-1
16	Use GPU	False
17	Log Experiment	False
18	Experiment Name	reg-default-name
19	USI	c230
20	Imputation Type	simple
21	Iterative Imputation Iteration	None
22	Numeric Imputer	median
23	Iterative Imputation Numeric Model	None

```
# create a model which gives the highest accuracy
lghm = create_model("lightgbm", fold=10)
```

	Fold	MAE	MSE	RMSE	R2	RMSLE	MAPE
CV-Val	0	0.3372	0.1680	0.4099	0.3278	0.2880	0.3342
	1	0.3357	0.1677	0.4095	0.3287	0.2886	0.3322
	2	0.3376	0.1683	0.4102	0.3267	0.2869	0.3385
	3	0.3386	0.1688	0.4108	0.3247	0.2879	0.3358
	4	0.3334	0.1645	0.4056	0.3420	0.2845	0.3330
	5	0.3350	0.1670	0.4087	0.3320	0.2861	0.3351
	6	0.3406	0.1708	0.4133	0.3167	0.2900	0.3348
	7	0.3376	0.1680	0.4099	0.3277	0.2863	0.3390
	8	0.3356	0.1664	0.4079	0.3343	0.2856	0.3365
	9	0.3371	0.1687	0.4108	0.3250	0.2890	0.3332
	Mean	0.3366	0.1676	0.4097	0.3286	0.2873	0.3352
	Std	0.0019	0.0016	0.0019	0.0063	0.0016	0.0022
Train	nan	0.3358	0.1668	0.4084	0.3328	0.2865	0.3341

```
tuned_lghm = tune_model(lghm, fold=5, optimize="RMSE")
```

	Fold	MAE	MSE	RMSE	R2	RMSLE	MAPE
CV-Val	0	0.3383	0.1682	0.4101	0.3270	0.2886	0.3344
	1	0.3400	0.1691	0.4112	0.3235	0.2878	0.3400
	2	0.3357	0.1664	0.4079	0.3344	0.2859	0.3357
	3	0.3407	0.1702	0.4125	0.3192	0.2887	0.3390
	4	0.3363	0.1681	0.4099	0.3279	0.2877	0.3373
	Mean	0.3382	0.1684	0.4103	0.3264	0.2877	0.3373

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Code git

[17]:
```
models()
```

[17]:

ID	Name	Reference	Turbo
lr	Linear Regression	cuml.linear_model.linear_regression.LinearRegr...	True
lasso	Lasso Regression	cuml.linear_model.lasso.Lasso	True
ridge	Ridge Regression	cuml.linear_model.ridge.Ridge	True
en	Elastic Net	cuml.linear_model.elastic_net.ElasticNet	True
lar	Least Angle Regression	sklearn.linear_model._least_angle.Lars	True
llar	Lasso Least Angle Regression	sklearn.linear_model._least_angle.LassoLars	True
omp	Orthogonal Matching Pursuit	sklearn.linear_model._omp.OrthogonalMatchingPu...	True
br	Bayesian Ridge	sklearn.linear_model._bayes.BayesianRidge	True
ard	Automatic Relevance Determination	sklearn.linear_model._bayes.ARDRegression	False
par	Passive Aggressive Regressor	sklearn.linear_model._passive_aggressive.Passi...	True
ransac	Random Sample Consensus	sklearn.linear_model._ransac.RANSACRegressor	False
tr	TheilSen Regressor	sklearn.linear_model._theil_sen.TheilSenRegressor	False
huber	Huber Regressor	sklearn.linear_model._huber.HuberRegressor	True
kr	Kernel Ridge	sklearn.kernel_ridge.KernelRidge	False
svm	Support Vector Regression	cuml.svm.svr.SVR	False
knn	K Neighbors Regressor	cuml.neighbors.kneighbors_regressor.KNeighbors...	True
dt	Decision Tree Regressor	sklearn.tree._classes.DecisionTreeRegressor	True
rf	Random Forest Regressor	cuml.ensemble.randomforestregressor.RandomFore...	True
et	Extra Trees Regressor	sklearn.ensemble._forest.ExtraTreesRegressor	True
ada	AdaBoost Regressor	sklearn.ensemble._weight_boosting.AdaBoostRegr...	True
gbr	Gradient Boosting Regressor	sklearn.ensemble._gb.GradientBoostingRegressor	True
mlp	MLP Regressor	sklearn.neural_network._multilayer_perceptron...	False

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Code | git

```
Baseline_model_exp01 = setup(
    data = df, target = 'app_complete_flag',
    session_id = 42, #fix_imbalance=False, #ignore_feature=[''],
    n_jobs=-1,use_gpu=True,
    log_experiment=True, experiment_name='Baseline_model_exp01',
    log_plots=True, log_data=True,
    silent=True, verbose=True,
    log_profile=False
)
```

	Description	Value
0	session_id	42
1	Target	app_complete_flag
2	Original Data	(215068, 12)
3	Missing Values	False
4	Numeric Features	7
5	Categorical Features	4
6	Ordinal Features	False
7	High Cardinality Features	False
8	High Cardinality Method	None
9	Transformed Train Set	(150547, 43)
10	Transformed Test Set	(64521, 43)
11	Shuffle Train-Test	True
12	Stratify Train-Test	False
13	Fold Generator	KFold
14	Fold Number	10
15	CPU Jobs	-1
16	Use GPU	True

lead_scoring... (13) - JupyterLab

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lead_scc ● | Terminal | Dummy | utils.py | constant | lead_sco | utils.py | constant | lead_scc | utils.py | constant | utils.py | Dummy | Python 3

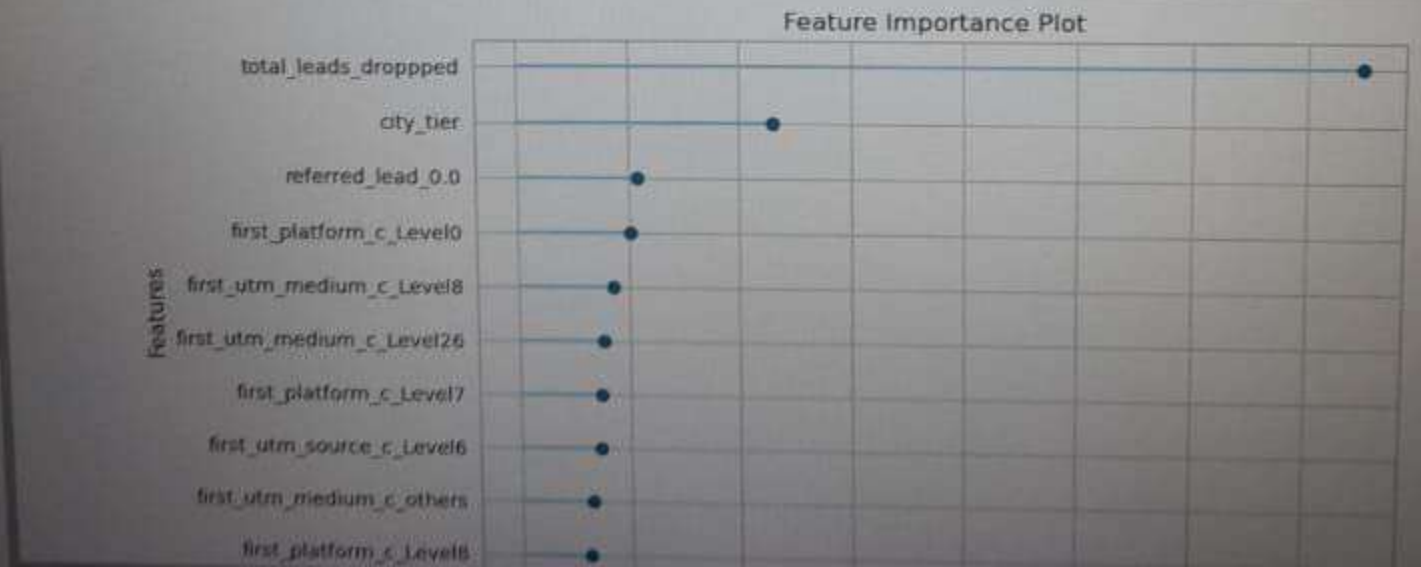
Code

[21]:
```
print(lgbm, "\n")
print(tuned_lgbm)
```

```
LGBMRegressor(boosting_type='gbdt', class_weight=None, colsample_bytree=1.0,
              device='gpu', importance_type='split', learning_rate=0.1,
              max_depth=-1, min_child_samples=20, min_child_weight=0.001,
              min_split_gain=0.0, n_estimators=100, n_jobs=-1, num_leaves=31,
              objective=None, random_state=42, reg_alpha=0.0, reg_lambda=0.0,
              silent='warn', subsample=1.0, subsample_for_bin=200000,
              subsample_freq=0)

LGBMRegressor(bagging_fraction=0.6, bagging_freq=3, boosting_type='gbdt',
              class_weight=None, colsample_bytree=1.0, device='gpu',
              feature_fraction=0.7, importance_type='split', learning_rate=0.2,
              max_depth=-1, min_child_samples=61, min_child_weight=0.001,
              min_split_gain=0.1, n_estimators=90, n_jobs=-1, num_leaves=40,
              objective=None, random_state=42, reg_alpha=0.0005, reg_lambda=0.7,
              silent='warn', subsample=1.0, subsample_for_bin=200000,
              subsample_freq=0)
```

[22]:
```
# create feature importance plot
plot_model(tuned_lgbm, plot="feature")
```



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Code git

4 : Model Experimentation with pycaret

[13]:
```
# create a experimentation with pycaret and exclude ['gbc','knn','qda', 'dummy', 'svm', 'ada']

data_unseen = df.sample(frac=0.1, random_state=42)      # Sample 10% of the data to become the unseen test set
df = df.drop(data_unseen.index)                         # Use the remaining 90% as the training (& validation) data

df.reset_index(drop=True, inplace=True)
data_unseen.reset_index(drop=True, inplace=True)

print('Data for Model Training & Validation: ' + str(df.shape))
print('Unseen Data For Predictions: ' + str(data_unseen.shape))
```

```
Data for Model Training & Validation: (215068, 12)
Unseen Data For Predictions: (23896, 12)
```

[14]:
```
df.head()
```

[14]:

	city_tier	first_platform_c	first_utm_medium_c	first_utm_source_c	total_leads_droppped	referred_lead	app_complete_flag	assistance_interaction	career_interaction	paymer
0	1.0	Level0	Level11	Level2	1.0	0.0	1	0.0	0.0	
1	2.0	Level3	Level0	others	1.0	0.0	1	0.0	0.0	
2	1.0	Level3	Level0	Level0	1.0	0.0	1	0.0	0.0	
3	1.0	Level1	Level3	others	2.0	0.0	0	0.0	0.0	
4	1.0	Level3	Level0	Level0	1.0	0.0	0	0.0	0.0	

[15]:
```
from pycaret.regression import *
```

[16]:
```
#expt_basic = setup(
#    data = df,
#    target = 'app_complete_flag',
#    session_id=42,                  # Random seed to ensure reproducibility of the experiment with the same data
#    train_size=0.8,                 # 80% training data & 20% held-out validation data
#    ignore_features=[],
#    numeric_imputation='median',    # "mean" by default
#    categorical_imputation='mode',  # "constant" (not available) by default
#)
```

5 : Model Experimentation after dropping features

From the above feature tests we can clearly see that some of the features are not significant. We will now drop all the insignificant features and select only the significant ones. The list of the significant features is ['total_leads_droppped', 'city_tier', 'referred_lead', 'app_complete_flag', 'first_platform_c', 'first_utm_medium_c', 'first_utm_source_c']. So now you will train your model with onlly these features.

Also note that in our previous expenments we saw that tree based models are our top performers. In case of tree based models we do not require transformationss normalization, scaling etc. So make sure that you use setup pycaret in the proper way. i.e, make sure that you use normalize = False and transformation = False.

[23]:
```
utils.build_dbs()
```

```
2.6.0
```

[24]:
```
utils.encode_features()
```

[25]:
```
utils.read_db_file(DB_FILE_NAME_NOTE)
```

```
[('features',), ('target',)]
```

[26]:
```
utils.read_table_data(FEAT_TAB_NAME)
```

```
Connected to features table
First row in Table:  (1.0, 'Level0', 'Level11', 'Level2', 1.0, 0.0, 1)
```

[27]:
```
utils.read_table_data(TARG_TAB_NAME)
```

```
Connected to target table
First row in Table:  (1,)
```

[28]:
```
#
# ['total_leads_droppped', 'city_tier', 'referred_lead', 'app_complete_flag', 'first_platform_c', 'first_utm_medium_c', 'first_utm_source_c']
#
# Train the model using the features listed above. Since we are using tree models we do not require any transformaions
# such as normalization, scaling etc.So make sure that you use setup pycaret in the proper way. i.e, make sure that you use
# normalize = False and transformation = False.

con = sqlite3.connect('/home/Assignment/02_training_pipeline/notebooks/lead_scoring_data_cleaning.db')
FEATDATA_FRAME = pd.read_sql("""SELECT * from features""", con)

Baseline_model_exp00 = setup(
    data = FEATDATA_FRAME, target = 'app_complete_flag',
    session_id = 42, fix_imbalance=False, #ignore_feature=['']
    n_jobs=-1,use_gpu=True,
    log_experiment=True, experiment_name='Baseline_model_exp00',
```

lead_scoring... (13) - JupyterLab

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Code git

[18]:
```
best = compare_models(sort="RMSE", exclude=['knn', 'dummy', 'svm', 'ada'], fold=5)
```

	Model	MAE	MSE	RMSE	R2	RMSLE	MAPE	TT (Sec)
lightgbm	Light Gradient Boosting Machine	0.3368	0.1678	0.4097	0.3286	0.2873	0.3352	4.0880
xgboost	Extreme Gradient Boosting	0.3354	0.1682	0.4101	0.3273	0.2874	0.3341	0.6220
rf	Random Forest Regressor	0.3350	0.1687	0.4108	0.3249	0.2878	0.3333	3.1440
et	Extra Trees Regressor	0.3338	0.1700	0.4123	0.3201	0.2887	0.3322	5.2520
dt	Decision Tree Regressor	0.3338	0.1703	0.4127	0.3185	0.2890	0.3322	0.2920
gbr	Gradient Boosting Regressor	0.3525	0.1720	0.4147	0.3118	0.2909	0.3510	8.8620
br	Bayesian Ridge	0.3891	0.1925	0.4387	0.2299	0.3060	0.3904	0.2660
ridge	Ridge Regression	0.3890	0.1925	0.4388	0.2299	0.3060	0.3903	0.0280
lr	Linear Regression	0.3917	0.1931	0.4394	0.2276	0.3066	0.3925	0.2380
huber	Huber Regressor	0.3611	0.2057	0.4535	0.1771	0.3141	0.3204	7.5760
omp	Orthogonal Matching Pursuit	0.4168	0.2075	0.4555	0.1700	0.3181	0.4165	0.0640
lasso	Lasso Regression	0.5000	0.2500	0.5000	-0.0002	0.3515	0.4975	0.0260
en	Elastic Net	0.5000	0.2500	0.5000	-0.0002	0.3515	0.4975	0.0420
llar	Lasso Least Angle Regression	0.5000	0.2500	0.5000	-0.0002	0.3515	0.4975	0.0620
par	Passive Aggressive Regressor	0.5133	0.6822	0.7769	-1.7291	0.4082	0.3815	0.3520
lar	Least Angle Regression	169.2187	734352.8107	479.1193	-2938196.6680	2.2495	160.9371	0.0720

[19]:
```
# create a model which gives the highest accuracy
lgbm = create_model("lightgbm", fold=10)
```

	MAE	MSE	RMSE	R2	RMSLE	MAPE

Experiments

Baseline_model_exp00

Track machine learning training runs in experiments. Learn more

Default

Baseline_model_exp00

Baseline_model_exp01

Experiment ID: 1

Description Edit

Refresh | Compare | Delete | Download CSV | Start Time | All time

Columns | Only show differences | metrics.rmse < 1 and params.model = "tree" | Search | Filter | Clear

Showing 11 matching runs

Start Time	Duration	Run Name	User	Source	Version	Models	Metrics > MAE	MAPE
1 hour ago		Session Initi...	root	ipykernel...	-	-		
1 hour ago		Linear Regre...	root	ipykernel...	-	-	-	-
1 hour ago		Ridge Regre...	root	ipykernel...	-	-	-	-
1 hour ago		Bayesian Rid...	root	ipykernel...	-	-	-	-
1 hour ago		Gradient Bo...	root	ipykernel...	-	sklearn	0.389	0.39
1 hour ago		Decision Tre...	root	ipykernel...	-	sklearn	0.389	0.39
1 hour ago		Extra Trees R...	root	ipykernel...	-	sklearn	0.353	0.351
1 hour ago		Random For...	root	ipykernel...	-	sklearn	0.334	0.332
1 hour ago		Extreme Gra...	root	ipykernel...	-	sklearn	0.334	0.332
1 hour ago		Light Gradie...	root	ipykernel...	-	sklearn	0.335	0.333
							0.335	0.334

Track machine learning training runs in experiments. Learn more

Experiment ID: 2

Description Edit

Refresh | Compare | Delete | Download CSV | Start Time | All time

Columns | Only show differences | metrics.rmse < 1 and params.model = "tree" | Search | Filter | Clear

Showing 19 matching runs

	Start Time	Duration	Run Name	User	Source	Version	Models	Metrics > MAE	MAPE	MSE
	1 hour ago		Session Initi...	root	ipykernel...	-	-	-	-	-
	57 minutes ago		Light Gradie...	root	ipykernel...	-	sklearn	0.339	0.337	0.168
	1 hour ago		Light Gradie...	root	ipykernel...	-	sklearn	0.337	0.335	0.168
	1 hour ago		Least Angle ...	root	ipykernel...	-	sklearn	47.19	42.7	35769.7
	1 hour ago		Passive Aggr...	root	ipykernel...	-	sklearn	0.513	0.382	0.682
	1 hour ago		Least Angle ...	root	ipykernel...	-	sklearn	0.5	0.498	0.25
	1 hour ago		Elastic Net	root	ipykernel...	-	sklearn	0.5	0.498	0.25
	1 hour ago		Lasso Regre...	root	ipykernel...	-	sklearn	0.5	0.498	0.25
	1 hour ago		Orthogonal ...	root	ipykernel...	-	sklearn	0.417	0.417	0.208
	1 hour ago		Huber Regre...	root	ipykernel...	-	sklearn	0.361	0.32	0.206
	1 hour ago		Linear Regre...	root	ipykernel...	-	sklearn	0.392	0.393	0.193
	1 hour ago		Ridge Regre...	root	ipykernel...	-	sklearn	0.389	0.39	0.193
	1 hour ago		Bayesian Rid...	root	ipykernel...	-	sklearn	0.389	0.39	0.193
	1 hour ago		Gradient Bo...	root	ipykernel...	-	sklearn	0.353	0.351	0.172
	1 hour ago		Decision Tre...	root	ipykernel...	-	sklearn			